Rapport détaillé — tuandromd copy (classification)

Pipeline live tfmv1: login SSH/API, libération d'un slot dataset, duplication du predefined tuandromd, cible Label, suppression group-optimizer des lignes ni malware ni goodware (normal), auto-clean-split, auto-transform, entraînement de tous les algorithmes de classification en hold-out ×10, avec documentation complète des logs, transformations, hyperparamètres, MV et résultats.

1. Contexte & identite du run

Plateforme	http://46.62.212.226 (tfmv1)
Utilisateur	cc@cc.cc
User ID	6a7649c85389ffb49703be24
Predefined source	tuandromd
Dataset copie	tuandromd_copy_20261026_160844
Colonne cible	Label
Labels conserves	goodware, malware (goodware = benign/normal)
Validation	holdout x 10
Debut (Madrid)	2026-10-26 17:08:43 CET
Fin (Madrid)	2026-10-26 17:10:06 CET
Debut (UTC)	2026-10-26T16:08:43.320339+00:00
Fin (UTC)	2026-10-26T16:10:06.513419+00:00

2. Gestion de la limite datasets (max 10)

Limite avant	{"count": 10, "limit_reached": true, "remaining": 0}
Datasets supprimes	[{"table_name": "hfj", "created_at": "2026-10-08T21:16:50.091026+00:00", "response": {"status": "success", "deleted_tables": ["hfj", "hfj_backup", "hfj_reset"], "message": "Successfully deleted 3 related tables"}}]
Limite apres delete	{"count": 9, "limit_reached": false, "remaining": 1}

Compte a 10/10. Suppression du plus ancien dataset principal (hors _backup/_reset/_numerized): hfj (cree 2026-10-08). Ensuite remaining=1, duplication autorisee.

3. Duplication predefined tuandromd

original_table_name	tuandromd
new_dataset_name	tuandromd_copy_20261026_160844
API	POST /api/duplicate-predefined-dataset
Reponse	{"status": "success", "message": "Dataset duplicated successfully", "main_table": "tuandromd_copy_20261026_160844", "backup_table": "tuandromd_copy_20261026_160844_backup", "reset_table": "tuandromd_copy_20261026_160844_reset", "columns": ["ACCESS_ALL_DOWNLOADS", "ACCESS_CACHE_FILESYSTEM", "ACCESS_CHECKIN_PROPERTIES", "ACCESS_COARSE_LOCATION", "ACCESS_COARSE_UPDATES", "ACCESS_FINE_LOCATION", "ACCESS_LOCATION_EXTRA_COMMANDS", "ACCESS_MOCK_LOCATION", "ACCESS_MTK_MMHW", "ACCESS_NETWORK_STATE", "ACCESS_PROVIDER", "ACCESS_SERVICE", "ACCESS_SHARED_DATA", "ACCESS_SUPERUSER", "ACCESS_SURFACE_FLINGER",

4. Target column

API	POST /api/save-target-column
dataset_name	tuandromd_copy_20261026_160844
target_column	Label
Note	Le dataset n'a pas de colonne Class; la cible reelle est Label.
Reponse	{"message": "Target column saved successfully!", "is_unsupervised": false}

5. Group Optimizer — delete-rows (ni malware ni goodware/normal)

Distribution initiale Label: malware=3565, goodware=899, [Empty]=1. Une seule ligne n'etait ni malware ni goodware (equivalent normal/benign). Suppression via POST /api/delete-rows (meme payload que Group Optimizer UI).

values_to_delete	["[Empty]"]
deleted_rows	1
Label apres	malware=3565, goodware=899 (total 4464)

6. Auto Clean Split — rapport & logs

API	POST /api/auto-clean-split/start (remove_duplicates=true)
Job ID	554c7031-5a23-497e-8893-d19c07d0c9e5
Status	done
Original	4464 rows x 242 cols, target=Label
Constant columns removed	42
Empty columns	[]
High-missing columns	[]
Empty rows	0
High-missing rows	0
Duplicate rows removed	3802
Target missing rows	0
Split	{"method": "stratified", "train_size": 0.8, "train_rows": 530, "test_rows": 132, "problem_type": "classification"}
Imputation (MV)	{"median": [], "mean": [], "mode": []}
Balance	{"ratio": 4.0476190476190474, "n_classes": 2, "n_train": 530, "n_min": 105, "minority_share": 0.19811320754716982, "under_share_threshold": 0.25, "post_under_size": 210, "method": "oversampling", "applied": true, "reason": "post_under_too_small"}
Validation flags	{"missing_train_features": 0, "missing_test_features": 0, "leakage_safe_imputation": true, "test_not_resampled": true}
Final sizes	{"rows": 982, "columns": 200, "train_rows": 850, "test_rows": 132}
Warnings	["Possible time-related columns: ACCESS_COARSE_UPDATES, CONTROL_LOCATION_UPDATES, SET_TIME, SET_TIME_ZONE. The split is random, not by time."]

6.1 Decision d'equilibrage (train)

Ratio majority/minority=4.048, n_classes=2, n_train=530, n_min=105, minority_share=0.198. Seuil under-share=0.25, post_under_size=210. Methode choisie: oversampling (applied=True, reason=post_under_too_small). Regle: undersampling si minority>=0.5/k ET k*n_min>=1000; sinon oversampling. Ici post_under trop petit (210 < 1000) => oversampling. Train apres resample: 850 rows; test non resampe: 132.

6.2 Colonnes constantes supprimees (42)

ACCESS_PROVIDER, ACCESS_SERVICE, ACCESS_SHARED_DATA, activityCalled, ACTIVITY_RECOGNITION, ANT, ANT_ADMIN, AUTORUN_MANAGER_LICENSE_MANAGER, AUTORUN_MANAGER_LICENSE_SERVICE_autorun, BIND_GET_INSTALL_REFERRER_SERVICE, CHANGE_DISPLAY_MODE, CHECK_LICENSE, DATABASE_INTERFACE_SERVICE, DOWNLOAD_SERVICE, EXTENSION_PERMISSION, FULLSCREEN_FULL, GOOGLE_AUTH, GOOGLE_PHOTOS, JPUSH_MESSAGE, MAPS_RECEIVE, MESSAGE, PERMISSION, PERMISSION_RUN_TASKS, PLUGIN, READ, READ_ATTACHMENT, READ_AVESTTINGS, READ_CONTENT_PROVIDER, READ_DATA, READ_DATABASES, READ_GMAIL, READ_GSERVICES, READ_MESSAGES, RECEIVE, RECEIVE_SIGNED_DATA_RESULT, RESPOND, REQUEST, SEND, WRITE, WRITE_AVSETTING, WRITE_DATA, WRITE_DATABASES

6.3 Logs auto-clean-split (complets)

```
[16:09:01] dataset: tuandromd_copy_20261026_160844
[16:09:01] 1. profile
[16:09:03] 4,464 rows x 242 columns
```

```
[16:09:03] missing cells: 0 (0.0%)
[16:09:03] target: Label
[16:09:03] duplicates: 3,802 (85.2%)
[16:09:03] problem type: classification
[16:09:03]  time-like  columns:  ACCESS_COARSE_UPDATES,  CONTROL_LOCATION_UPDATES,  SET_TIME,  SET_TIME_ZONE  (split  is  not
chronological)
[16:09:03] 2. clean
[16:09:03] drop column ACCESS_PROVIDER: constant (1 unique value)
[16:09:03] drop column ACCESS_SERVICE: constant (1 unique value)
[16:09:03] drop column ACCESS_SHARED_DATA: constant (1 unique value)
[16:09:03] drop column activityCalled: constant (1 unique value)
[16:09:03] drop column ACTIVITY_RECOGNITION: constant (1 unique value)
[16:09:03] drop column ANT: constant (1 unique value)
[16:09:03] drop column ANT_ADMIN: constant (1 unique value)
[16:09:03] drop column AUTORUN_MANAGER_LICENSE_MANAGER: constant (1 unique value)
[16:09:03] drop column AUTORUN_MANAGER_LICENSE_SERVICE_autorun: constant (1 unique value)
[16:09:03] drop column BIND_GET_INSTALL_REFERRER_SERVICE: constant (1 unique value)
[16:09:03] drop column CHANGE_DISPLAY_MODE: constant (1 unique value)
[16:09:03] drop column CHECK_LICENSE: constant (1 unique value)
[16:09:03] drop column DATABASE_INTERFACE_SERVICE: constant (1 unique value)
[16:09:03] drop column DOWNLOAD_SERVICE: constant (1 unique value)
[16:09:03] drop column EXTENSION_PERMISSION: constant (1 unique value)
[16:09:03] drop column FULLSCREEN_FULL: constant (1 unique value)
[16:09:03] drop column GOOGLE_AUTH: constant (1 unique value)
[16:09:03] drop column GOOGLE_PHOTOS: constant (1 unique value)
[16:09:03] drop column JPUSH_MESSAGE: constant (1 unique value)
[16:09:03] drop column MAPS_RECEIVE: constant (1 unique value)
[16:09:03] drop column MESSAGE: constant (1 unique value)
[16:09:03] drop column PERMISSION: constant (1 unique value)
[16:09:03] drop column PERMISSION_RUN_TASKS: constant (1 unique value)
[16:09:03] drop column PLUGIN: constant (1 unique value)
[16:09:03] drop column READ: constant (1 unique value)
[16:09:03] drop column READ_ATTACHMENT: constant (1 unique value)
[16:09:03] drop column READ_AVESTTINGS: constant (1 unique value)
[16:09:03] drop column READ_CONTENT_PROVIDER: constant (1 unique value)
[16:09:03] drop column READ_DATA: constant (1 unique value)
[16:09:03] drop column READ_DATABASES: constant (1 unique value)
[16:09:03] drop column READ_GMAIL: constant (1 unique value)
[16:09:03] drop column READ_GSERVICES: constant (1 unique value)
[16:09:03] drop column READ_MESSAGES: constant (1 unique value)
[16:09:03] drop column RECEIVE: constant (1 unique value)
[16:09:03] drop column RECEIVE_SIGNED_DATA_RESULT: constant (1 unique value)
[16:09:03] drop column RESPOND: constant (1 unique value)
[16:09:03] drop column REQUEST: constant (1 unique value)
[16:09:03] drop column SEND: constant (1 unique value)
[16:09:03] drop column WRITE: constant (1 unique value)
[16:09:03] drop column WRITE_AVSETTING: constant (1 unique value)
[16:09:03] drop column WRITE_DATA: constant (1 unique value)
[16:09:03] drop column WRITE_DATABASES: constant (1 unique value)
[16:09:04] no empty target rows
[16:09:05] no empty or almost-empty rows
[16:09:05] drop 3,802 duplicates
[16:09:05] 3. split
[16:09:05] method: stratified (same class mix in train and test)
[16:09:05] train 80% / test 20% (662 rows)
[16:09:05] result: train 530 / test 132
[16:09:05] 4. fill missing (stats from train only)
[16:09:06] no missing features, skip fill
[16:09:06] 5. class balance (train only)
[16:09:06] train classes: goodware=425, malware=105
[16:09:06] max/min ratio: 4.0:1 | minority_share=0.198 | under_threshold=0.250 | post_under≈210
[16:09:06] mild imbalance → oversample train (reason=post_under_too_small)
[16:09:06] oversampling on train. test unchanged
[16:09:06] 6. check
[16:09:07] missing feature cells: train 0, test 0
[16:09:07] train and test features have no missing cells
[16:09:07] fill values came from train only
[16:09:07] test was not resampled
[16:09:07] done.
```

7. Auto Transform — decisions, MV, logs

API	POST /api/auto-transform/start
Job ID	09443b97-d711-44ac-9a6f-6cc45327f804
Status	done
scaling_strategy	none
date_strategy	keep
kept_date_column	None
Transforms applied	[{"column": "Label", "method": "binary", "ok": true}]
Deleted columns	[]
Empty rows deleted	0
Left unchanged (count)	199
Warnings	[]

7.1 Transformation cible Label

Seule transformation appliquee: Label → Binary Encoding (malware/goodware → 0.0/1.0). Toutes les features boolean deja en 0/1: leave (skip Binary Encoding). Aucune imputation MV supplementaire (missing train/test features = 0 apres clean). scaling_strategy=none (defaut UI Auto Transform).

7.2 Decisions par colonne (resume)

leave/None	199
transform/binary	1

Decision Label: {"column": "Label", "role": "target", "action": "transform", "method": "binary", "confidence": 0.95, "reasons": ["Boolean target → Binary Encoding (true/false → 1/0)"]}

7.3 Logs auto-transform (echantillon + fin)

```
[16:09:11] dataset: tuandromd_copy_20261026_160844
[16:09:11] extract from text first if you need fields from free text
[16:09:11] scaling: none | dates: keep
[16:09:11] 1. profile
[16:09:12] 982 rows x 200 columns
[16:09:12] target: Label (supervised)
[16:09:12] 2. decide
[16:09:12] ACCESS_ALL_DOWNLOADS [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_CACHE_FILESYSTEM [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_CHECKIN_PROPERTIES [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_COARSE_LOCATION [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_COARSE_UPDATES [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_FINE_LOCATION [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_LOCATION_EXTRA_COMMANDS [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_MOCK_LOCATION [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_MTK_MMHW [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_NETWORK_STATE [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_SUPERUSER [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_SURFACE_FLINGER [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCESS_WIFI_STATE [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ACCOUNT_MANAGER [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] ADD_VOICEMAIL [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
```

```
[16:09:12] AUTHENTICATE_ACCOUNTS [boolean] leave  score=95%
[16:09:12] Already 0/1 , skip Binary Encoding
[16:09:12] BATTERY_STATS [boolean] leave  score=95%
...
[16:09:13] Lcom_google_android_gms_location_LocationClient_getLastLocation [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_location_LocationManager_getLastKgoodwarewnLocation [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getDeviceId [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_content_pm_PackageManager_getInstalledApplications [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_content_pm_PackageManager_getInstalledPackages [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getLine1Number [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getNetworkOperator [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getNetworkOperatorName [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getNetworkCountryIso [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getSimOperator [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getSimOperatorName [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getSimCountryIso [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Landroid_telephony_TelephonyManager_getSimSerialNumber [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Lorg_apache_http_impl_client_DefaultHttpClient_execute [boolean] leave  score=95%
[16:09:13] Already 0/1 , skip Binary Encoding
[16:09:13] Label [target] transform/binary  score=95%
[16:09:13] Boolean target to Binary Encoding (true/false to 1/0)
[16:09:13] 3. drop columns
[16:09:13] nothing to drop
[16:09:13] 4. apply
[16:09:13] 1 transform(s) in one pass: binary:Label
[16:09:13] full table rewrite once, wait if the dataset is large
[16:09:14] batch applied (1 transform(s))
[16:09:14] Label → Binary encoding
[16:09:14] 5. drop empty rows
[16:09:14] drop 0 empty rows
[16:09:14] done.
```

Total lignes de log auto-transform: 417 (artifact JSON complet).

8. Entrainements classification — hold-out x10

Six algorithmes entraines sequentiellement sur le dataset apres clean+transform. Validation: Hold-Out Repeated x10 (test_size=0.2 sur le train plateforme). Metriques finales UI = evaluation sur le test set global (132 lignes).

8.1 Tableau recapitulatif

1. Logistic Regression | holdout 0.9541±0.0090 | test acc=0.9470 prec=0.9504 rec=0.9470 f1=0.9480 auc=0.9829 | duree=5.0s | ok=True

2. k-Nearest Neighbors (KNN) | holdout 0.9288±0.0225 | test acc=0.9394 prec=0.9417 rec=0.9394 f1=0.9402 auc=0.9717 | duree=5.1s | ok=True

3. LightGBM | holdout 0.9718±0.0073 | test acc=0.9545 prec=0.9545 rec=0.9545 f1=0.9545 auc=0.9913 | duree=5.0s | ok=True

4. Random Forest | holdout 0.9776±0.0111 | test acc=0.9621 prec=0.9620 rec=0.9621 f1=0.9612 auc=— | duree=10.1s | ok=True

5. Neural Net — MLP (PyTorch) | holdout 0.9547±0.0151 | test acc=0.9318 prec=0.9441 rec=0.9318 f1=0.9347 auc=0.9877 | duree=20.1s | ok=True

6. Support Vector Machine (SVM) | holdout 0.9500±0.0185 | test acc=0.9697 prec=0.9712 rec=0.9697 f1=0.9701 auc=— | duree=5.0s | ok=True

9.1 Logistic Regression

Identite & timing

Algorithm	logistic
Training ID	logistic_1793030956_182084e0
Dataset	tuandromd_copy_20261026_160844
User ID	6a7649c85389ffb49703be24
Status	success
Started (UTC)	2026-10-26T16:09:16.180558+00:00
Started (Madrid)	2026-10-26 17:09:16 CET
Finished (UTC)	2026-10-26T16:09:20.296112+00:00
Finished (Madrid)	2026-10-26 17:09:20 CET
Duration (logger)	4.118 s
Duration (runner)	5.039 s

Hyperparametres / request (start log)

C_search_space	None
bootstrap_params	None
class_weight	None
cv_params	None
date_column	None
l1_ratio	None
max_iter	1000
model_type	logistic
multi_class	auto
n_repetitions	10
n_trials	20
numerized_dataset_name	tuandromd_copy_20261026_160844
original_dataset_name	tuandromd_copy_20261026_160844
penalty	l2
random_search	False
random_search_seed	42
random_state	None
solver	lbfgs
time_series_params	None
training_id	logistic_1793030956_182084e0
use_scaling	True
validation_method	holdout

Hyperparametres effectifs (end log)

metadata	{"model_type": "logistic_regression", "C_used": 1.0, "C_selection_method": "validation_split", "penalty": "l2", "solver": "lbfgs", "multi_class": "auto", "n_features": 199, "n_train_samples": 850, "n_test_samples": 132, "validation_method": "holdout", "class_names": ["0.0", "1.0"], "used_validation_model": true, "final_model_source": "best_holdout_repetition_7", "date_column_excluded": false}
features_used	[199 features]
best_C_used	1.0
C_search_space	[0.001, 0.01, 0.1, 1, 10, 100, 1000]
C_selection_method	validation_split
training_info	{"model_trained_once": false, "used_cv_best_model": false, "used_bootstrap_model": false, "used_holdout_model": true, "used_timeseries_model": false, "final_model_source": "best_holdout_repetition_7", "training_samples": 850, "test_samples": 132,

"parameter_optimization": "C optimization using holdout", "n_repetitions": 10, "validation_method": "holdout", "date_column_excluded": false, "date_colum

Methode de validation (MV hold-out)

Methode	Hold-Out Repeated (10 repetitions)
n_repetitions	10
test_size interne	0.2
mean_score	0.9541
std_score	0.0090
best_repetition	7
best_score	0.9706
cv_train_mean	0.9828
cv_train_std	0.0121

Scores par repetition

rep 1: val=0.9529 train=0.9926 sizes=680/170 best_C=10.0

rep 2: val=0.9529 train=0.9765 sizes=680/170 best_C=1.0

rep 3: val=0.9529 train=0.9971 sizes=680/170 best_C=1000.0

rep 4: val=0.9529 train=0.9691 sizes=680/170 best_C=0.1

rep 5: val=0.9353 train=0.9882 sizes=680/170 best_C=10.0

rep 6: val=0.9529 train=0.9662 sizes=680/170 best_C=0.1

rep 7: val=0.9706 train=0.9721 sizes=680/170 best_C=1.0

rep 8: val=0.9588 train=0.9971 sizes=680/170 best_C=1000.0

rep 9: val=0.9647 train=0.9721 sizes=680/170 best_C=1.0

rep 10: val=0.9471 train=0.9971 sizes=680/170 best_C=1000.0

Resultats test set global

accuracy	0.9470
precision	0.9504
recall	0.9470
f1	0.9480
roc_auc	0.9829
train_accuracy	0.9718
log_loss	0.2050
n_classes	2
classes	["0.0", "1.0"]
confusion_matrix	{"true_negative": 101, "false_positive": 5, "false_negative": 2, "true_positive": 24, "labels": ["0.0", "1.0"]}

Classification report


```
{
  "0.0": {
    "precision": 0.9805825242718447,
    "recall": 0.9528301886792453,
    "f1-score": 0.9665071770334929,
    "support": 106.0
  },
  "1.0": {
    "precision": 0.8275862068965517,
    "recall": 0.9230769230769231,
    "f1-score": 0.8727272727272727,
    "support": 26.0
  },
  "accuracy": 0.946969696969697,
  "macro avg": {
```

```
    "precision": 0.9040843655841981,
    "recall": 0.9379535558780843,
    "f1-score": 0.9196172248803828,
    "support": 132.0
  },
  "weighted avg": {
    "precision": 0.9504468860009536,
    "recall": 0.946969696969697,
    "f1-score": 0.9480353777004494,
    "support": 132.0
  }
}
```


Top feature importance (15)

SYSTEM_ALERT_WINDOW: 1.370639

BATTERY_STATS: 1.266981

READ_CONTACTS: -1.216817

BLUETOOTH: -1.091555

WRITE_SETTINGS: -1.044733

Landroid_location_LocationManager_getLastKgoodwarewnLocation: -1.038451

VIBRATE: -1.010123

MANAGE_ACCOUNTS: 0.981555

RECEIVE_BOOT_COMPLETED: 0.978212

RECEIVE_SMS: 0.931998

Landroid_telephony_TelephonyManager_getSimOperator: 0.926231

CHANGE_NETWORK_STATE: -0.900462

GET_TASKS: 0.893414

SEND_SMS: 0.877696

READ_EXTERNAL_STORAGE: 0.858128

9.2 k-Nearest Neighbors (KNN)

Identite & timing

Algorithm	knn
Training ID	knn_1793030961_3cbb4bdf
Dataset	tuandromd_copy_20261026_160844
User ID	6a7649c85389ffb49703be24
Status	success
Started (UTC)	2026-10-26T16:09:21.223164+00:00
Started (Madrid)	2026-10-26 17:09:21 CET
Finished (UTC)	2026-10-26T16:09:24.549819+00:00
Finished (Madrid)	2026-10-26 17:09:24 CET
Duration (logger)	3.327 s
Duration (runner)	5.057 s

Hyperparametres / request (start log)

algorithm	auto
bootstrap_params	None
cv_params	None
final_training_on_all_data	False
leaf_size	30
metric	minkowski
model_type	knn
n_neighbors	5
n_repetitions	10
n_trials	20
numerized_dataset_name	tuandromd_copy_20261026_160844
original_dataset_name	tuandromd_copy_20261026_160844
p	2
random_search	False
random_search_seed	42
time_series_column	None
time_series_params	None
training_id	knn_1793030961_3cbb4bdf
use_scaling	False
validation_method	holdout
weights	uniform

Hyperparametres effectifs (end log)

n_neighbors_used	5
weights_used	uniform
metric_used	minkowski
algorithm_used	auto
features_used	[199 features]
final_params_used	{"n_neighbors": 5, "weights": "uniform", "algorithm": "auto", "leaf_size": 30, "p": 2, "metric": "minkowski"}
training_info	{"final_training_data": "train_only", "all_data_samples": 850, "train_samples_used": 850, "test_samples_used": 0, "final_model_trained_on": "train_data_only", "n_repetitions": 10, "validation_method": "holdout", "date_column_excluded": false, "scaling_applied": false, "model_trained_once": false, "used_cv_best_model": false, "used_bootstrap_model": false, "used_holdout_repeated_model": true, "used

Methode de validation (MV hold-out)

Methode	Hold-Out Repeated (10 repetitions)
n_repetitions	10
test_size interne	0.2
mean_score	0.9288
std_score	0.0225
best_repetition	3
best_score	0.9529
cv_train_mean	0.9479
cv_train_std	0.0085

Scores par repetition

rep 1: val=0.9471 train=0.9485 sizes=680/170

rep 2: val=0.8941 train=0.9632 sizes=680/170

rep 3: val=0.9529 train=0.9485 sizes=680/170

rep 4: val=0.9529 train=0.9397 sizes=680/170

rep 5: val=0.9294 train=0.9456 sizes=680/170

rep 6: val=0.8824 train=0.9471 sizes=680/170

rep 7: val=0.9412 train=0.9529 sizes=680/170

rep 8: val=0.9294 train=0.9426 sizes=680/170

rep 9: val=0.9353 train=0.9588 sizes=680/170

rep 10: val=0.9235 train=0.9324 sizes=680/170

Resultats test set global

accuracy	0.9394
precision	0.9417
recall	0.9394
f1	0.9402
roc_auc	0.9717
train_accuracy	0.9647
log_loss	—
n_classes	2
classes	["0.0", "1.0"]
confusion_matrix	{"true_negative": 101, "false_positive": 5, "false_negative": 3, "true_positive": 23, "labels": ["0.0", "1.0"]}

Classification report


```
{
 "0.0": {
  "precision": 0.9711538461538461,
  "recall": 0.9528301886792453,
  "f1-score": 0.9619047619047619,
  "support": 106.0
 },
 "1.0": {
  "precision": 0.8214285714285714,
  "recall": 0.8846153846153846,
  "f1-score": 0.8518518518518519,
  "support": 26.0
 },
```

```
  "accuracy": 0.9393939393939394,
  "macro avg": {
    "precision": 0.8962912087912087,
    "recall": 0.9187227866473149,
    "f1-score": 0.9068783068783068,
    "support": 132.0
  },
  "weighted avg": {
    "precision": 0.9416625041625042,
    "recall": 0.9393939393939394,
    "f1-score": 0.9402276735610069,
    "support": 132.0
  }
}
```


Top feature importance (15)

Landroid_telephony_TelephonyManager_getDeviceId: 0.011912

Ljava_net_URL_openConnection: 0.011898

Ljavax_crypto_Cipher_doFinal: 0.011870

READ_PHONE_STATE: 0.011801

RECEIVE_BOOT_COMPLETED: 0.011758

Landroid_telephony_TelephonyManager_getNetworkOperator: 0.011612

WRITE_EXTERNAL_STORAGE: 0.011507

GET_TASKS: 0.011313

WAKE_LOCK: 0.011205

Ljava_lang_reflect_Method_invoke: 0.011174

Ljava_lang_Runtime_exec: 0.011110

ACCESS_WIFI_STATE: 0.011099

Landroid_telephony_TelephonyManager_getNetworkOperatorName: 0.011088

Landroid_location_LocationManager_getLastKgoodwarewnLocation: 0.011077

ACCESS_NETWORK_STATE: 0.010799

9.3 LightGBM

Identite & timing

Algorithm	lightgbm
Training ID	lightgbm_1793030966_6e2addad
Dataset	tuandromd_copy_20261026_160844
User ID	6a7649c85389ffb49703be24
Status	success
Started (UTC)	2026-10-26T16:09:26.285661+00:00
Started (Madrid)	2026-10-26 17:09:26 CET
Finished (UTC)	2026-10-26T16:09:29.706021+00:00
Finished (Madrid)	2026-10-26 17:09:29 CET
Duration (logger)	3.420 s
Duration (runner)	5.043 s

Hyperparametres / request (start log)

boosting_type	gbdt
bootstrap_params	None
colsample_bytree	1.0
cv_params	None
early_stopping_rounds	10
final_training_on_all_data	False
learning_rate	0.1
max_depth	-1
min_child_samples	20
model_type	lightgbm
n_estimators	100
n_jobs	-1
n_repetitions	10
n_trials	20
num_leaves	31
numerized_dataset_name	tuandromd_copy_20261026_160844
objective	None
original_dataset_name	tuandromd_copy_20261026_160844
random_search	False
random_search_seed	42
random_state	42
reg_alpha	0.0
reg_lambda	0.0
subsample	1.0
time_series_column	None
time_series_params	None
training_id	lightgbm_1793030966_6e2addad
validation_method	holdout
verbose	-1

Hyperparametres effectifs (end log)

objective_used	binary
boosting_type_used	gbdt
n_estimators_used	100
learning_rate_used	0.1
features_used	[199 features]

final_params_used	{"boosting_type": "gbdt", "num_leaves": 31, "learning_rate": 0.1, "n_estimators": 100, "max_depth": -1, "min_child_samples": 20, "subsample": 1.0, "colsample_bytree": 1.0, "reg_alpha": 0.0, "reg_lambda": 0.0, "random_state": 42, "n_jobs": -1, "verbose": -1, "objective": "binary", "early_stopping_rounds": 10}
training_info	{"final_training_data": "train_only", "all_data_samples": 850, "train_samples_used": 850, "test_samples_used": 0, "final_model_trained_on": "train_data_only", "n_repetitions": 10, "validation_method": "holdout", "date_column_excluded": false, "early_stopping_used": true, "best_iteration": 0, "model_trained_once": false, "used_cv_best_model": false, "used_bootstrap_model": false, "used_holdout_repe

Methode de validation (MV hold-out)

Methode	Hold-Out Repeated (10 repetitions)
n_repetitions	10
test_size interne	0.2
mean_score	0.9718
std_score	0.0073
best_repetition	8
best_score	0.9824
cv_train_mean	0.9959
cv_train_std	0.0024

Scores par repetition

rep 1: val=0.9765 train=0.9971 sizes=680/170 best_iter=99

rep 2: val=0.9706 train=0.9956 sizes=680/170 best_iter=100

rep 3: val=0.9706 train=0.9971 sizes=680/170 best_iter=90

rep 4: val=0.9765 train=0.9926 sizes=680/170 best_iter=100

rep 5: val=0.9529 train=0.9971 sizes=680/170 best_iter=99

rep 6: val=0.9706 train=0.9985 sizes=680/170 best_iter=95

rep 7: val=0.9706 train=0.9971 sizes=680/170 best_iter=96

rep 8: val=0.9824 train=0.9912 sizes=680/170 best_iter=79

rep 9: val=0.9765 train=0.9985 sizes=680/170 best_iter=98

rep 10: val=0.9706 train=0.9941 sizes=680/170 best_iter=91

Resultats test set global

accuracy	0.9545
precision	0.9545
recall	0.9545
f1	0.9545
roc_auc	0.9913
train_accuracy	0.9976
log_loss	—
n_classes	2
classes	["0.0", "1.0"]
confusion_matrix	{"true_negative": 103, "false_positive": 3, "false_negative": 3, "true_positive": 23, "labels": ["0.0", "1.0"]}

Classification report

{
  "0.0": {
    "precision": 0.9716981132075472,
    "recall": 0.9716981132075472,

```
    "f1-score": 0.9716981132075472,
    "support": 106.0
  },
  "1.0": {
    "precision": 0.8846153846153846,
    "recall": 0.8846153846153846,
    "f1-score": 0.8846153846153846,
    "support": 26.0
  },
  "accuracy": 0.9545454545454546,
  "macro avg": {
    "precision": 0.9281567489114659,
    "recall": 0.9281567489114659,
    "f1-score": 0.9281567489114659,
    "support": 132.0
  },
  "weighted avg": {
    "precision": 0.9545454545454546,
    "recall": 0.9545454545454546,
    "f1-score": 0.9545454545454546,
    "support": 132.0
  }
}
```


Top feature importance (15)

Ljava_lang_Runtime_exec: 0.059847

READ_PHONE_STATE: 0.051496

WRITE_EXTERNAL_STORAGE: 0.051148

RECEIVE_BOOT_COMPLETED: 0.048017

Ljava_net_URL_openConnection: 0.046277

WAKE_LOCK: 0.043841

Ljavax_crypto_Cipher_doFinal: 0.039318

INTERNET: 0.035143

ACCESS_NETWORK_STATE: 0.034099

Ljava_lang_reflect_Method_invoke: 0.033751

Landroid_telephony_TelephonyManager_getDeviceId: 0.028184

Landroid_content_pm_PackageManager_getInstalledPackages: 0.028184

SYSTEM_ALERT_WINDOW: 0.027488

KILL_BACKGROUND_PROCESSES: 0.026444

VIBRATE: 0.025400

9.4 Random Forest

Identite & timing

Algorithm	random_forest
Training ID	random_forest_1793030971_a32d063c
Dataset	tuandromd_copy_20261026_160844
User ID	6a7649c85389ffb49703be24
Status	success
Started (UTC)	2026-10-26T16:09:31.329441+00:00
Started (Madrid)	2026-10-26 17:09:31 CET
Finished (UTC)	2026-10-26T16:09:37.264074+00:00
Finished (Madrid)	2026-10-26 17:09:37 CET
Duration (logger)	5.935 s
Duration (runner)	10.064 s

Hyperparametres / request (start log)

bootstrap	True
bootstrap_params	None
criterion	gini
cv_params	None
final_training_on_all_data	False
max_depth	None
max_features	sqrt
min_samples_leaf	1
min_samples_split	2
model_type	random_forest
n_estimators	100
n_jobs	-1
n_repetitions	10
n_trials	20
numerized_dataset_name	tuandromd_copy_20261026_160844
oob_score	False
original_dataset_name	tuandromd_copy_20261026_160844
random_search	False
random_search_seed	42
time_series_column	None
time_series_params	None
training_id	random_forest_1793030971_a32d063c
validation_method	holdout

Hyperparametres effectifs (end log)

features_used	[199 features]
final_params_used	{"n_estimators": 100, "max_depth": null, "min_samples_split": 2, "min_samples_leaf": 1, "max_features": "sqrt", "bootstrap": true, "oob_score": false, "n_jobs": -1, "random_state": null, "criterion": "gini"}
user_parameters_used	True
training_info	{"final_training_data": "train_only", "all_data_samples": 850, "train_samples_used": 850, "test_samples_used": 0, "final_model_trained_on": "train_data_only", "n_repetitions": 10, "validation_method": "holdout", "date_column_excluded": false, "final_training_on_all_data": false, "training_samples": 850, "test_samples": 132, "final_model_samples": 850}

Methode de validation (MV hold-out)

Methode	Hold-Out Repeated (10 repetitions)
n_repetitions	10
test_size interne	0.2
mean_score	0.9776
std_score	0.0111
best_repetition	3
best_score	0.9941
cv_train_mean	0.9974
cv_train_std	0.0006

Scores par repetition

rep 1: val=0.9529 train=0.9971 sizes=680/170

rep 2: val=0.9824 train=0.9985 sizes=680/170

rep 3: val=0.9941 train=0.9971 sizes=680/170

rep 4: val=0.9706 train=0.9971 sizes=680/170

rep 5: val=0.9882 train=0.9971 sizes=680/170

rep 6: val=0.9765 train=0.9971 sizes=680/170

rep 7: val=0.9765 train=0.9971 sizes=680/170

rep 8: val=0.9882 train=0.9985 sizes=680/170

rep 9: val=0.9765 train=0.9971 sizes=680/170

rep 10: val=0.9706 train=0.9971 sizes=680/170

Resultats test set global

accuracy	0.9621
precision	0.9620
recall	0.9621
f1	0.9612
roc_auc	—
train_accuracy	0.9976
log_loss	—
n_classes	2
classes	["0.0", "1.0"]
confusion_matrix	{"true_negative": 105, "false_positive": 1, "false_negative": 4, "true_positive": 22, "labels": ["0.0", "1.0"]}

Classification report


```
{
  "0.0": {
    "precision": 0.963302752293578,
    "recall": 0.9905660377358491,
    "f1-score": 0.9767441860465116,
    "support": 106.0
  },
  "1.0": {
    "precision": 0.9565217391304348,
    "recall": 0.8461538461538461,
    "f1-score": 0.8979591836734694,
    "support": 26.0
  },
  "accuracy": 0.9621212121212122,
  "macro avg": {
```

```
    "precision": 0.9599122457120064,
    "recall": 0.9183599419448476,
    "f1-score": 0.9373516848599905,
    "support": 132.0
  },
  "weighted avg": {
    "precision": 0.9619670981856862,
    "recall": 0.9621212121212122,
    "f1-score": 0.9612259280033366,
    "support": 132.0
  }
}
```


Top feature importance (15)

RECEIVE_BOOT_COMPLETED: 0.110727

Ljava_net_URL_openConnection: 0.052456

RECEIVE_SMS: 0.046463

GET_TASKS: 0.046143

READ_PHONE_STATE: 0.045328

KILL_BACKGROUND_PROCESSES: 0.042330

SYSTEM_ALERT_WINDOW: 0.032407

Landroid_location_LocationManager_getLastKgoodwarewnLocation: 0.031847

WAKE_LOCK: 0.029028

VIBRATE: 0.028964

SEND_SMS: 0.025810

BATTERY_STATS: 0.025709

Ljava_lang_Runtime_exec: 0.022815

Landroid_telephony_TelephonyManager_getDeviceId: 0.021264

WRITE_EXTERNAL_STORAGE: 0.019408

9.5 Neural Net — MLP (PyTorch)

Identite & timing

Algorithm	mlp
Training ID	mlp_1793030981_1fabcac8
Dataset	tuandromd_copy_20261026_160844
User ID	6a7649c85389ffb49703be24
Status	success
Started (UTC)	2026-10-26T16:09:41.395431+00:00
Started (Madrid)	2026-10-26 17:09:41 CET
Finished (UTC)	2026-10-26T16:09:59.406250+00:00
Finished (Madrid)	2026-10-26 17:09:59 CET
Duration (logger)	18.011 s
Duration (runner)	20.074 s

Hyperparametres / request (start log)

activation	relu
batch_size	32
bootstrap_params	None
cv_params	None
dropout_rate	0.2
epochs	100
layers	[64, 32]
learning_rate	0.001
model_type	mlp
n_repetitions	10
numerized_dataset_name	tuandromd_copy_20261026_160844
original_dataset_name	tuandromd_copy_20261026_160844
patience	10
problem_type	auto
time_series_column	None
time_series_params	None
training_id	mlp_1793030981_1fabcac8
use_scaling	True
validation_method	holdout
validation_split	0.2

Hyperparametres effectifs (end log)

layers_used	[64, 32]
activation_used	relu
dropout_rate_used	0.2
learning_rate_used	0.001
batch_size_used	32
training_info	{"model_trained_once": false, "used_cv_best_model": false, "used_bootstrap_model": false, "used_holdout_model": true, "used_timeseries_model": false, "training_samples": 850, "test_samples": 132, "parameter_optimization": "User parameters used - no optimization", "final_model_source": "best_holdout_repetition", "early_stopping_used": true, "training_id": "mlp_1793030981_1fabcac8", "date_column_exc

Methode de validation (MV hold-out)

Methode	Hold-Out Repeated (10 repetitions)
n_repetitions	10

test_size interne	0.2
mean_score	0.9547
std_score	0.0151
best_repetition	1
best_score	0.9824
cv_train_mean	0.9547
cv_train_std	0.0151

Scores par repetition

rep 1: val=0.9824 train=— sizes=680/170

rep 2: val=0.9529 train=— sizes=680/170

rep 3: val=0.9529 train=— sizes=680/170

rep 4: val=0.9588 train=— sizes=680/170

rep 5: val=0.9353 train=— sizes=680/170

rep 6: val=0.9647 train=— sizes=680/170

rep 7: val=0.9706 train=— sizes=680/170

rep 8: val=0.9353 train=— sizes=680/170

rep 9: val=0.9353 train=— sizes=680/170

rep 10: val=0.9588 train=— sizes=680/170

Resultats test set global

accuracy	0.9318
precision	0.9441
recall	0.9318
f1	0.9347
roc_auc	0.9877
train_accuracy	0.9953
log_loss	—
n_classes	2
classes	["0.0", "1.0"]
confusion_matrix	{"true_negative": 98, "false_positive": 8, "false_negative": 1, "true_positive": 25, "labels": ["0.0", "1.0"]}

Classification report


```
{
  "0.0": {
    "precision": 0.98989898989899,
    "recall": 0.9245283018867925,
    "f1-score": 0.9560975609756097,
    "support": 106.0
  },
  "1.0": {
    "precision": 0.7575757575757576,
    "recall": 0.9615384615384616,
    "f1-score": 0.847457627118644,
    "support": 26.0
  },
  "accuracy": 0.9318181818181818,
  "macro avg": {
    "precision": 0.8737373737373737,
    "recall": 0.9430333817126271,
```

```
    "f1-score": 0.9017775940471269,
    "support": 132.0
  },
  "weighted avg": {
    "precision": 0.9441383532292622,
    "recall": 0.9318181818181818,
    "f1-score": 0.9346987861249954,
    "support": 132.0
  }
}
```

9.6 Support Vector Machine (SVM)

Identite & timing

Algorithm	svm
Training ID	svm_1793031001_13109b2c
Dataset	tuandromd_copy_20261026_160844
User ID	6a7649c85389ffb49703be24
Status	success
Started (UTC)	2026-10-26T16:10:01.472590+00:00
Started (Madrid)	2026-10-26 17:10:01 CET
Finished (UTC)	2026-10-26T16:10:06.217337+00:00
Finished (Madrid)	2026-10-26 17:10:06 CET
Duration (logger)	4.745 s
Duration (runner)	5.040 s

Hyperparametres / request (start log)

C	1.0
bootstrap_params	None
cv_params	None
degree	3
final_training_on_all_data	False
gamma	scale
kernel	rbf
model_type	svm
n_repetitions	10
n_trials	20
numerized_dataset_name	tuandromd_copy_20261026_160844
original_dataset_name	tuandromd_copy_20261026_160844
probability	False
random_search	False
random_search_seed	42
random_state	None
shrinking	True
time_series_column	None
time_series_params	None
training_id	svm_1793031001_13109b2c
validation_method	holdout

Hyperparametres effectifs (end log)

kernel_used	rbf
C_used	1.0
gamma_used	scale
degree_used	3
features_used	[199 features]
final_params_used	{"kernel": "rbf", "C": 1.0, "gamma": "scale", "degree": 3, "shrinking": true, "probability": false}
user_parameters_used	True
training_info	{"final_training_data": "train_only", "all_data_samples": 850, "train_samples_used": 850, "test_samples_used": 0, "final_model_trained_on": "train_data_only", "n_repetitions": 10, "validation_method": "holdout", "date_column_excluded": false, "final_training_on_all_data": false, "training_samples": 850, "test_samples": 132, "final_model_samples": 850, "feature_scaling": "StandardScaler applied", "

Methode de validation (MV hold-out)

Methode	Hold-Out Repeated (10 repetitions)
n_repetitions	10
test_size interne	0.2
mean_score	0.9500
std_score	0.0185
best_repetition	4
best_score	0.9706
cv_train_mean	0.9660
cv_train_std	0.0050

Scores par repetition

rep 1: val=0.9588 train=0.9618 sizes=680/170

rep 2: val=0.9647 train=0.9632 sizes=680/170

rep 3: val=0.9471 train=0.9662 sizes=680/170

rep 4: val=0.9706 train=0.9676 sizes=680/170

rep 5: val=0.9294 train=0.9735 sizes=680/170

rep 6: val=0.9412 train=0.9647 sizes=680/170

rep 7: val=0.9706 train=0.9706 sizes=680/170

rep 8: val=0.9647 train=0.9735 sizes=680/170

rep 9: val=0.9412 train=0.9603 sizes=680/170

rep 10: val=0.9118 train=0.9588 sizes=680/170

Resultats test set global

accuracy	0.9697
precision	0.9712
recall	0.9697
f1	0.9701
roc_auc	—
train_accuracy	0.9729
log_loss	—
n_classes	2
classes	["0.0", "1.0"]
confusion_matrix	{"true_negative": 103, "false_positive": 3, "false_negative": 1, "true_positive": 25, "labels": ["0.0", "1.0"]}

Classification report


```
{
 "0.0": {
  "precision": 0.9903846153846154,
  "recall": 0.9716981132075472,
  "f1-score": 0.9809523809523809,
  "support": 106.0
 },
 "1.0": {
  "precision": 0.8928571428571429,
  "recall": 0.9615384615384616,
  "f1-score": 0.9259259259259259,
  "support": 26.0
 },
```

```
  "accuracy": 0.9696969696969697,
  "macro avg": {
    "precision": 0.9416208791208791,
    "recall": 0.9666182873730044,
    "f1-score": 0.9534391534391534,
    "support": 132.0
  },
  "weighted avg": {
    "precision": 0.9711746586746587,
    "recall": 0.9696969696969697,
    "f1-score": 0.9701138367805034,
    "support": 132.0
  }
}
```


Top feature importance (15)

RECEIVE_BOOT_COMPLETED: 52.807006

BATTERY_STATS: 47.036843

RECEIVE_SMS: 44.793098

SYSTEM_ALERT_WINDOW: 39.594035

RECEIVE_WAP_PUSH: 37.714718

SEND_SMS: 37.297270

KILL_BACKGROUND_PROCESSES: 33.710139

READ_PHONE_STATE: 33.377649

USES_POLICY_FORCE_LOCK: 32.017152

READ_CALL_LOG: 31.719707

VIBRATE: 31.108301

MOUNT_FORMAT_FILESYSTEMS: 30.915423

GET_TASKS: 30.250415

ADD_VOICEMAIL: 28.884473

WRITE_CALL_LOG: 28.884473

10. Conclusion

Meilleure accuracy test: Support Vector Machine (SVM) = 0.9697 (F1=0.9701). Dataset final: 850 train / 132 test, 200 colonnes. Tous les algorithmes de classification ont ete entraines avec hold-out x10. Artifacts JSON complets (logs clean/transform, start/end training) archives sous backend/training_logs/tuandromd_pipeline/ et reports/tuandromd_pipeline/.

Fichiers sources

```
pipeline_artifact.json — etapes + logs clean/transform
final_results.json — resultats complets par algo
run_summary.jsonl / runner.log / console.log
*_start_*.json / *_end_*.json — training_logger
```